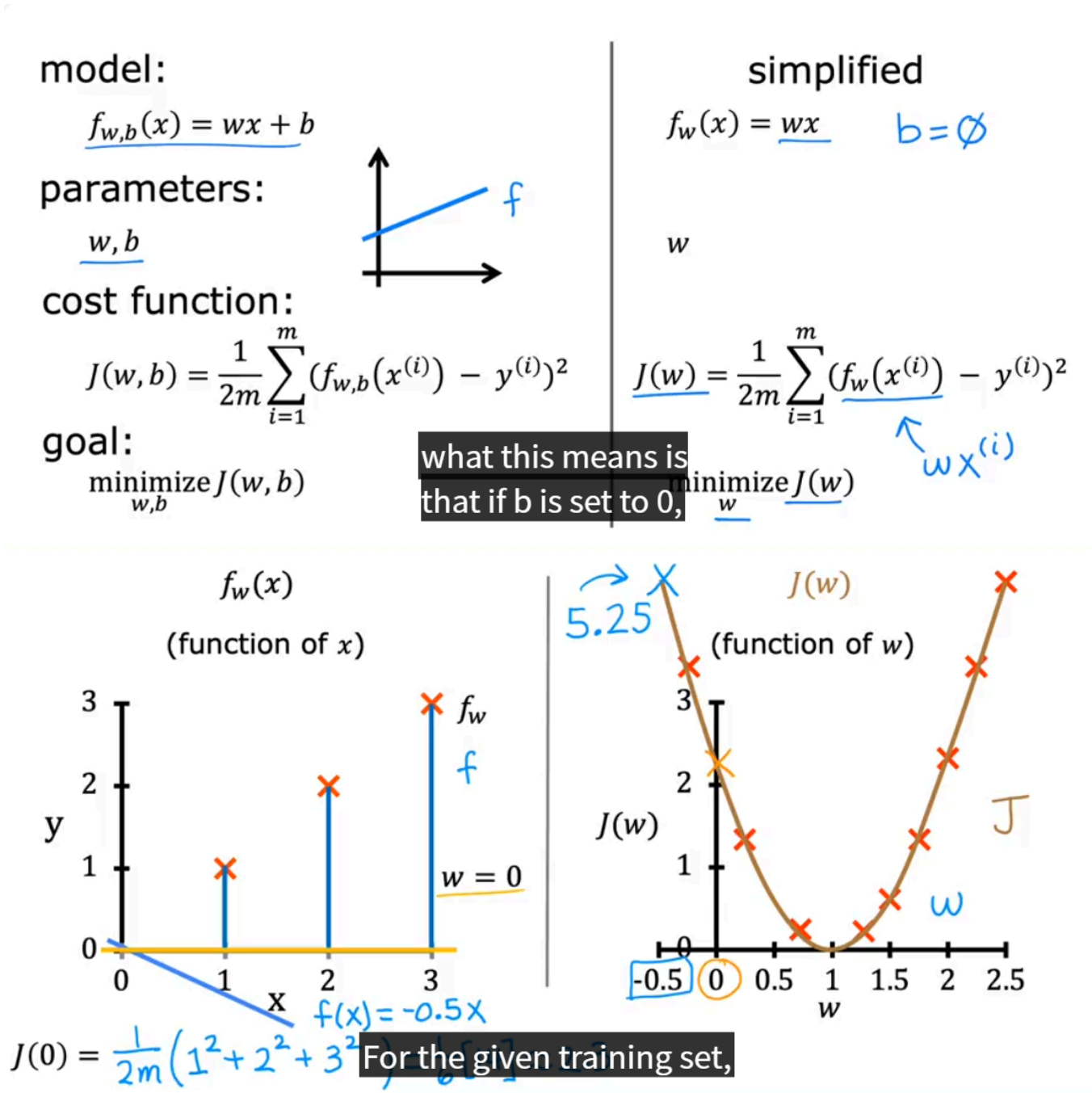
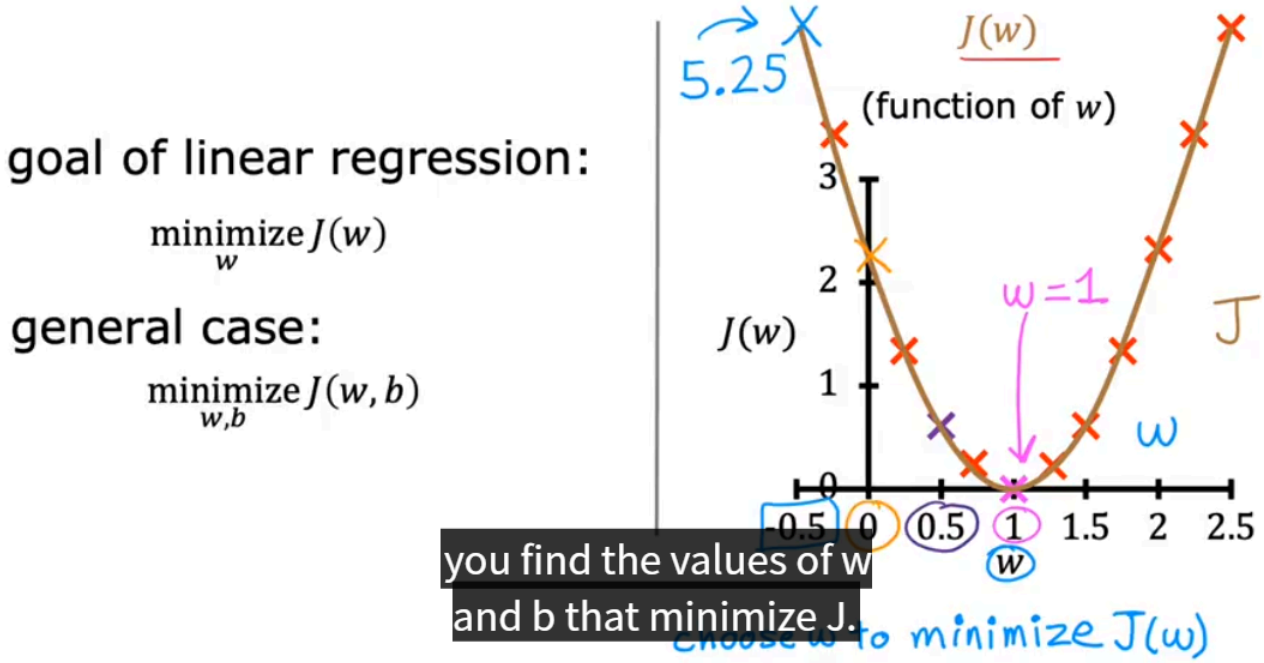
Day 3.Regression model & cost function

Cost function intuition

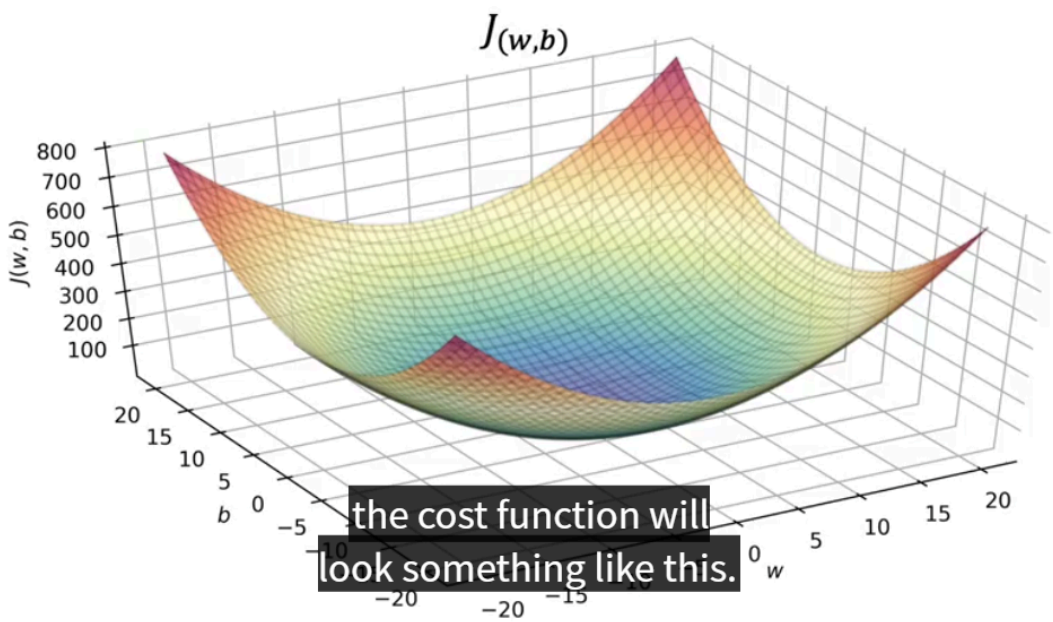




visualizing the cost function

things we seen so far

- model
- parameters
- cost function
- objective



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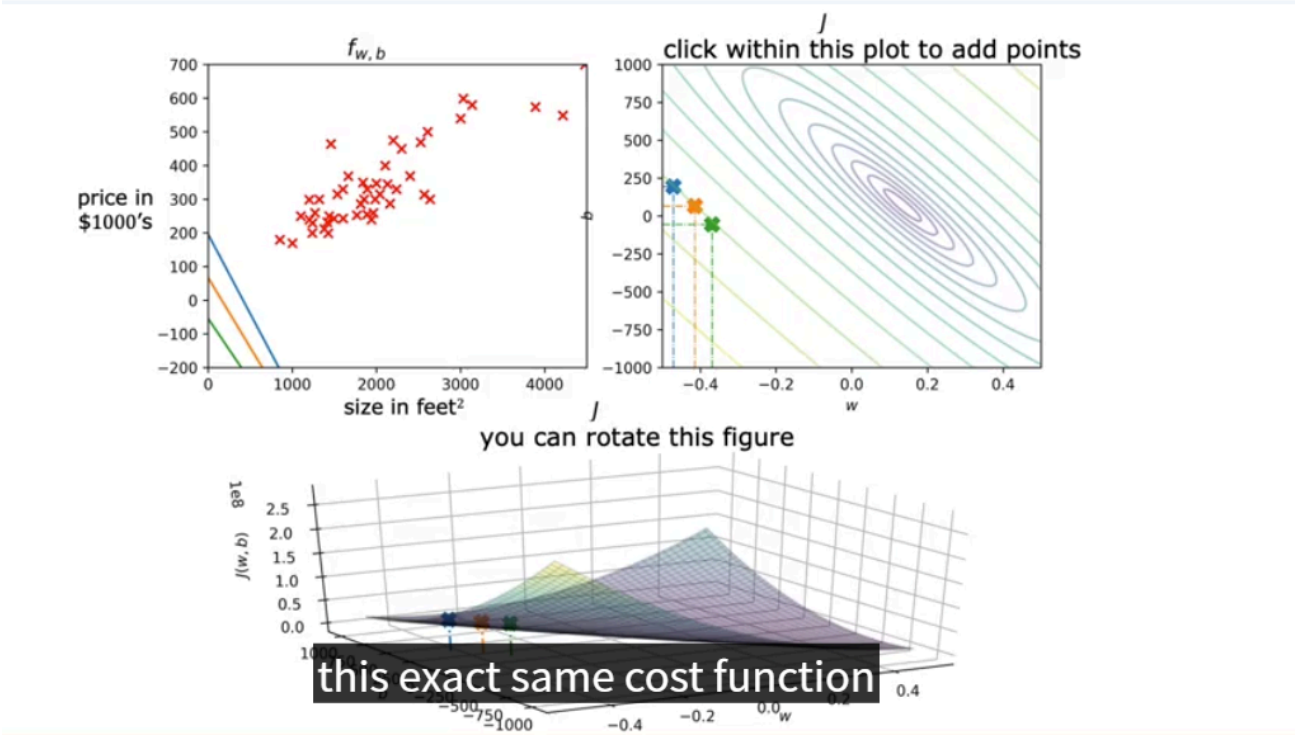
there is some better way to visualize the lost function.

like contour plots.



It is not wise to find the best practices of the parameters manually, in fact, we want to find a algorithm that can help us find the best parameters automatically, which is gradient descend. We will talk about this later.

more on cost function (optional lab)

```
def compute_cost(x, y, w, b):
    """
    Computes the cost function for linear regression.

    Args:
      x (ndarray (m,)): Data, m examples
      y (ndarray (m,)): target values
      w,b (scalar)    : model parameters

    Returns
        total_cost (float): The cost of using w,b as the parameters for linear
regression
               to fit the data points in x and y
    """
    # number of training examples
    m = x.shape[0]

    cost_sum = 0
    for i in range(m):
```

```
        f_wb = w * x[i] + b
        cost = (f_wb - y[i]) ** 2
        cost_sum = cost_sum + cost
    total_cost = (1 / (2 * m)) * cost_sum

    return total_cost
```